

AI-Powered Product Review Authenticity and Fake Review Detection System

PHASE I REPORT

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BONAFIDE CERTIFICATE

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“This project addresses the following Sustainable Development Goal(s): **Goal 9: Industry, Innovation and Infrastructure.**”

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ABSTRACT

Online product reviews strongly influence purchasing decisions, yet the review ecosystem now faces two different kinds of manipulation: deceptive reviews written by paid or incentivised people, and reviews generated by large language models. Most existing detectors treat both as one “fake” class, rely on the text alone and report accuracy on data drawn from the same source they were trained on. This project presents an AI-powered product review authenticity and fake review detection system that examines a review through three independent signals. A deceptive-writing model trained on the gold-standard Ott et al. corpus and an AI-generated-text model trained on GPT-2 and GLM-5.2 product reviews both use word and character TF-IDF features with logistic regression. A rule-based reviewer-behaviour module detects posting bursts, high posting velocity, duplicate and near-duplicate reviews and extreme rating patterns. The available signals are fused into an explainable Authenticity Score from 0 to 100 that shows the evidence behind every deduction and lists the checks that could not be run. The system is implemented as a FastAPI service with a React dashboard. On held-out data, the deception model reached 90.9% accuracy (ROC-AUC 0.972) on hotel reviews, and the AI-text model detected 91.7% of GPT-2 and 97.3% of GLM-5.2 reviews and 64% of a blind ChatGPT set never used in training, while flagging 6.3% of genuine reviews. The fused score achieved a ROC-AUC of 0.952 on 1,645 evaluation reviews. The remaining limitations are measured and reported openly.

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
API	Application Programming Interface
AUC	Area Under the Curve
CORS	Cross-Origin Resource Sharing
CSV	Comma-Separated Values
CV	Cross-Validation
DFD	Data Flow Diagram
ER	Entity–Relationship
FN / FP	False Negative / False Positive
FNR / FPR	False-Negative Rate / False-Positive Rate
GLM	General Language Model
GPT	Generative Pre-trained Transformer
HTTP	Hypertext Transfer Protocol
JSON	JavaScript Object Notation
LLM	Large Language Model
NLP	Natural Language Processing
ORM	Object–Relational Mapping
REST	Representational State Transfer
ROC	Receiver Operating Characteristic
SDG	Sustainable Development Goal
SHA	Secure Hash Algorithm
SPA	Single-Page Application
TF-IDF	Term Frequency–Inverse Document Frequency
TN / TP	True Negative / True Positive
UI	User Interface
UML	Unified Modeling Language

CHAPTER 1

INTRODUCTION

1.1 GENERAL

Customer reviews have become one of the most influential sources of information in electronic commerce. Because reviews shape the visibility and sales of a product, they also attract manipulation. Sellers can pay people to post glowing reviews of their own products or damaging reviews of competitors, and a study of a large review platform found that about 16 percent of restaurant reviews were filtered as suspicious, with businesses under competitive pressure more likely to commit review fraud [1].

The nature of the problem has changed in recent years. Early fake reviews were written by people, often recruited through crowdsourcing platforms, and their language differs measurably from genuine experience: deceptive writers tend to describe what they imagine rather than what they lived through [2]. Large language models now make it possible to produce fluent, varied and on-topic reviews at almost no cost, and recent work reports that human readers are close to chance when asked to separate real product reviews from reviews written by such models [3]. A fake review may therefore be written by a paid person, generated by a machine, or be a copied text posted by a suspicious account, and each case leaves different evidence.

Text is not the only evidence. Accounts that post fake reviews often behave unusually: they post many reviews within a short burst, repeat the same text, or give almost only one-star and five-star ratings [4], [5], [6]. Such behaviour can reveal a suspicious reviewer even when every individual review reads naturally.

This project, titled *AI-Powered Product Review Authenticity and Fake Review Detection System*, builds a web platform that detects fake product reviews using natural language processing and machine learning, analyses reviewer behaviour, computes a review authenticity score and visualises the results through a dashboard. Instead of producing a bare yes-or-no label, the system combines three independent signals into an

explainable score and states which checks could and could not be carried out for each review. By supporting trustworthy information in online marketplaces, the project contributes to Sustainable Development Goal 9, Industry, Innovation and Infrastructure.

1.2 OBJECTIVE

The main objective of the project is to design, implement and rigorously evaluate a system that estimates how much confidence a reader can place in an online product review. The specific objectives are the following.

- To detect human-written deceptive reviews from their language, using a model trained on a gold-standard corpus of verified fake reviews.
- To detect machine-generated review text, and to evaluate this capability on a text generator that was never seen during training.
- To analyse reviewer behaviour, including posting bursts, posting velocity, exact and near-duplicate reviews and rating patterns, and to express it as a reviewer risk score with explicit reasons.
- To fuse the available signals into a single Review Authenticity Score from 0 to 100 with per-signal attribution, plain-language findings and an explicit list of the checks that could not be run.
- To provide a REST API and a web dashboard for checking reviews, importing review data, exploring reviewer risk and monitoring analyses.
- To validate every component on held-out data with leakage controls, and to document the limitations that remain.

1.3 EXISTING SYSTEM

Existing approaches to fake review detection fall into a few families. Review platforms rely on manual moderation, user reports and proprietary filters whose criteria are not disclosed. Academic systems usually train a supervised text classifier, such as n-gram features with a support vector machine or a fine-tuned transformer, on one labelled corpus and report its accuracy. Behaviour-based and graph-based methods model reviewer activity and reviewer–product networks instead of text [7], [8], and more recent hybrid models fuse text and behaviour features in a single classifier [9].

Although these systems report high accuracy on their own benchmarks, they share several drawbacks when applied to present-day e-commerce reviews.

- **A single notion of fake.** Human deception and machine generation are treated as one class, although they leave different linguistic traces and require different training data.
- **In-distribution evaluation.** Models are commonly tested on data from the same source or text generator used for training, which overstates their performance on reviews from new language models.
- **Opaque verdicts.** Many systems output a label or a probability without explaining which evidence produced it.
- **No notion of missing evidence.** When reviewer history is unavailable, most systems either ignore behaviour silently or implicitly treat the absence of history as a clean record.
- **A verdict for every input.** Very short or out-of-domain reviews are scored with the same apparent confidence as long, in-domain ones.

1.4 PROPOSED SYSTEM

The proposed system assesses every review with three independent signals and fuses them into one explainable Review Authenticity Score. Figure 1.1 gives an overview.

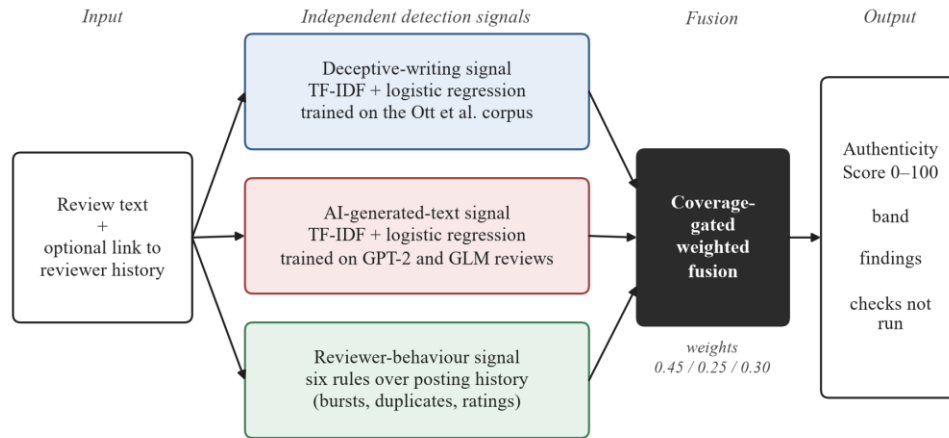


Figure 1.1. Overview of the proposed three-signal review authenticity system

The **deceptive-writing signal** estimates the probability that a review was written by a person paid to fake it. It is a logistic regression classifier over word and character TF-IDF features, trained on the Ott et al. deceptive opinion spam corpus, whose deceptive half was written by crowdworkers instructed to invent convincing reviews [2], [10]. The **AI-generated-text signal** estimates the probability that the text was produced by a language model. It uses the same feature pipeline but is trained on product reviews generated by GPT-2 [11] and GLM-5.2 together with real Amazon reviews [12], [13], and it is evaluated on ChatGPT-written reviews that were never used for training. The **reviewer-behaviour signal** applies six weighted rules to the posting history of a reviewer and produces a risk score from 0 to 100 with a reason for every rule that fires.

The signals are combined by coverage-gated weighted fusion: only the signals that could actually be evaluated contribute, their weights are renormalised, and a clean behaviour profile cannot raise the score by more than five points. The score is mapped to three bands, *authentic*, *uncertain* and *suspicious*, and is returned together with the findings behind it and the checks that could not be performed.

Compared with the existing approaches, the proposed system separates human-written deception from AI-generated text as attributable signals, validates its AI-text detector on a generator withheld from training, explains every score, never counts a missing signal as evidence of authenticity, and reports its known failure cases alongside the headline results.

CHAPTER 2

LITERATURE SURVEY

2.1 LITERATURE REVIEW

Research on fake review detection spans more than fifteen years and draws on natural language processing, behavioural analytics and, most recently, the detection of machine-generated text. This section reviews representative work in four groups: detection of human-written deceptive reviews, reviewer-behaviour and network methods, detection of machine-generated reviews, and hybrid, explainable and selective approaches.

2.1.1 HUMAN-WRITTEN DECEPTION DETECTION

Jindal and Liu [4] carried out one of the first large-scale studies of opinion spam, using reviews from Amazon. Because no verified fake reviews were available, they treated duplicate and near-duplicate reviews as spam and trained a logistic regression model over review, reviewer and product features. The study showed that opinion spam is widespread and that duplication is a practical signal, although the labels were a proxy rather than confirmed deception.

Ott et al. [2] created the first gold-standard corpus of deceptive opinion spam by asking crowdworkers on Amazon Mechanical Turk to write positive reviews of Chicago hotels they had not visited, and paired them with truthful reviews of the same hotels. Human judges performed close to chance on this data, whereas n-gram features with a support vector machine separated the classes with close to 90 percent accuracy. The deceptive reviews contained more imaginative and less spatial language, confirming that deception leaves detectable traces. Ott, Cardie and Hancock [10] later added 400 deceptive and 400 truthful negative reviews, producing the balanced corpus of 1,600 reviews that is used for the deceptive-writing model of this project.

Mukherjee et al. [14] studied reviews removed by the commercial filter of Yelp and found that linguistic n-gram models were much less accurate on these real filtered

reviews than on crowdsourced fake reviews, while behavioural features of the reviewers were far more discriminative. The result warns that models trained on crowdsourced deception do not necessarily transfer to real platforms. Cross-domain studies on the Ott corpus and related restaurant and doctor reviews report that accuracy near 86 percent within one domain falls to roughly 52 to 64 percent across domains, and domain adaptation methods [15] try to recover part of this loss by aligning the representations of different domains.

2.1.2 BEHAVIOUR AND NETWORK-BASED DETECTION

Lim et al. [5] modelled the rating behaviour of spammers, such as concentrating on particular products and deviating from the ratings of other reviewers, and ranked reviewers by a combined spam score. Mukherjee, Liu and Glance [16] moved from individual spammers to groups, mining sets of reviewers who reviewed the same products together and ranking the groups with behavioural indicators such as a narrow time window and rating deviation.

Fei et al. [6] observed that fake reviews tend to arrive in bursts and combined burst detection with a Markov random field over reviewers to identify spammers active during those bursts, and a burst-based unsupervised method was later proposed for detecting spammer groups [17]. Rayana and Akoglu [7] introduced SpEagle, which combines text and behavioural metadata with the review network in one framework, together with the widely used YelpChi, YelpNYC and YelpZip datasets. Graph neural network approaches [8] have since learned reviewer and product representations directly from such networks, gaining accuracy at the cost of interpretability.

Luca and Zervas [1] analysed review fraud from an economic perspective. They found that about 16 percent of restaurant reviews on Yelp were filtered, and that businesses facing more competition or a worsening reputation were more likely to engage in fraud, which underlines both the scale of the problem and the incentives behind it.

2.1.3 MACHINE-GENERATED REVIEW DETECTION

Yao et al. [18] showed that a character-level recurrent neural network trained on Yelp reviews could generate fake reviews that human evaluators rarely recognised and even rated as useful, and proposed a defence based on differences between the character

distributions of generated and genuine text. Adelani et al. [19] used GPT-2 [11] to generate sentiment-preserving fake reviews that fooled human readers, and found that machine-based detectors identified them more reliably than people.

Salminen et al. [12] generated 20,216 fake product reviews with GPT-2 across ten Amazon product categories, paired them with the same number of real reviews and fine-tuned a RoBERTa-based detector, fakeRoBERTa, which detected the generated reviews far more reliably than human judges. Their dataset supplies the GPT-2 and human classes of the AI-text model in this project. General-purpose detectors of machine-generated text include the RoBERTa-based GPT-2 output detector released by OpenAI [20] and zero-shot methods such as DetectGPT [21], which compares the log-probability of a passage with that of perturbed versions and requires access to a scoring language model. After the release of ChatGPT, Guo et al. [22] published a comparison corpus of human and ChatGPT answers and studied the differences between them and methods for detecting ChatGPT text.

Newer datasets and studies focus on reviews written by large language models. MAiDE-up [23] provides 10,000 real and 10,000 GPT-generated hotel reviews in ten languages. A 2025 study [24] fine-tuned DistilBERT to separate ChatGPT-rephrased reviews from real ones and reported 97.25 percent accuracy with LIME and SHAP explanations, evaluated on a balanced dataset built with the same generation process as its training data. Meng et al. [3] found that people distinguished real from LLM-generated product reviews with 50.8 percent accuracy, essentially chance, and that LLM judges did no better. Studies of cross-generator generalisation for AI-generated text and code [25] show that detectors often lose accuracy when the generator at test time differs from the generators seen in training.

2.1.4 HYBRID AND SELECTIVE APPROACHES

WF-CFRB [9] fuses contextual text features with reviewer-behaviour features through weighted fusion, using a convolutional network over behaviour vectors, and is the closest architectural analogue to the system proposed here; it does not address machine-generated text. An explainable hybrid deep learning model for e-commerce platforms [26] combines RoBERTa embeddings, TF-IDF features, a long short-term

memory network and multi-head attention, and explains its predictions with SHAP values and attention visualisation. More generally, model-agnostic explanation techniques such as LIME [27] and SHAP [28] are widely used to justify individual predictions of text classifiers.

Selective prediction, in which a classifier may decline to decide, has a long history, from the reject option analysed by Chow [29] to deep networks with an integrated reject option such as SelectiveNet [30]. In the literature surveyed, however, no fake review detector was found that abstains on inputs carrying too little evidence or that refuses to apply a model outside the domain on which it was validated.

2.2 COMPARISON AND DISCUSSION

Table 2.1 compares representative approaches along the capabilities that matter for present-day review authenticity. A dash marks a property that does not apply, and entries marked not verified could not be confirmed from the accessible parts of the publication.

Table 2.1. Comparison of surveyed review authenticity approaches

Approach	Domain	Human deception	AI text	Behaviour	Signal fusion	Explanation	Unseen generation or test
Ott et al. [2], [10]	Hotel	Yes	No	No	No	Partial	–
Mukherjee et al. [14]	Yelp	Yes	No	Yes	Partial	Partial	–
SpEagle, GNN [7], [8]	Yelp	Partial	No	Yes	Yes	No	–
Burst methods [6], [17]	Reviews	No	No	Yes	No	Partial	–
Salminen et al. [12]	Products	No	Yes	No	No	No	No
MAiDE-up [23]	Hotel	Partial	Yes	No	No	Partial	Not verified
DistilBERT study [24]	Products	No	Yes	No	No	Yes	No
WF-CFRB [9]	Reviews	Yes	No	Yes	Yes	Not verified	–
Hybrid XAI model [26]	E-commerce	Yes	No	Not verified	Yes	Yes	–
Proposed system	Products	Yes (hotel-validated)	Yes	Yes	Yes	Yes	Yes (ChatGPT)

Three observations follow from the comparison. First, text-based deception detection, behaviour analysis, the fusion of text with behaviour and post-hoc explanation have each been studied before, and none of them is claimed as new in this project. Second, human-written deception and machine-generated text are almost always handled by separate lines of work, and the fused systems surveyed treat a fake review as a single class. Third, most detectors of AI-written reviews are evaluated on text from the same generator that produced their training data. The 97.25 percent reported for ChatGPT-rephrased reviews [24] and the 64 percent measured in this project on unseen ChatGPT reviews therefore measure different things and should not be compared directly.

Accuracy figures across the literature are also not comparable because the datasets, domains and evaluation protocols differ. For this reason the contribution of this project is framed in terms of capability and evaluation methodology rather than as a claim of state-of-the-art accuracy.

2.3 CONCLUSION

The survey shows that the ingredients of review authenticity analysis are individually well established but rarely combined in a way that reflects the present threat. In the literature surveyed, no single product review system was found that separates human-written deception from AI-generated text as independent signals, adds reviewer behaviour as a third independent signal, fuses the signals into one explainable score with per-signal attribution that never treats missing evidence as clean, and validates its AI-text detector on a generator withheld from training. This project addresses that gap. Because the review was targeted rather than exhaustive, the gap is described as rare in the surveyed literature rather than as a first-ever contribution.

CHAPTER 3

SYSTEM DESIGN

3.1 SYSTEM ARCHITECTURE

The platform follows the layered client–server architecture of Figure 3.1: a React single-page application communicates with a FastAPI back end through a versioned REST API, while model training and evaluation run offline and hand over serialised model artifacts.

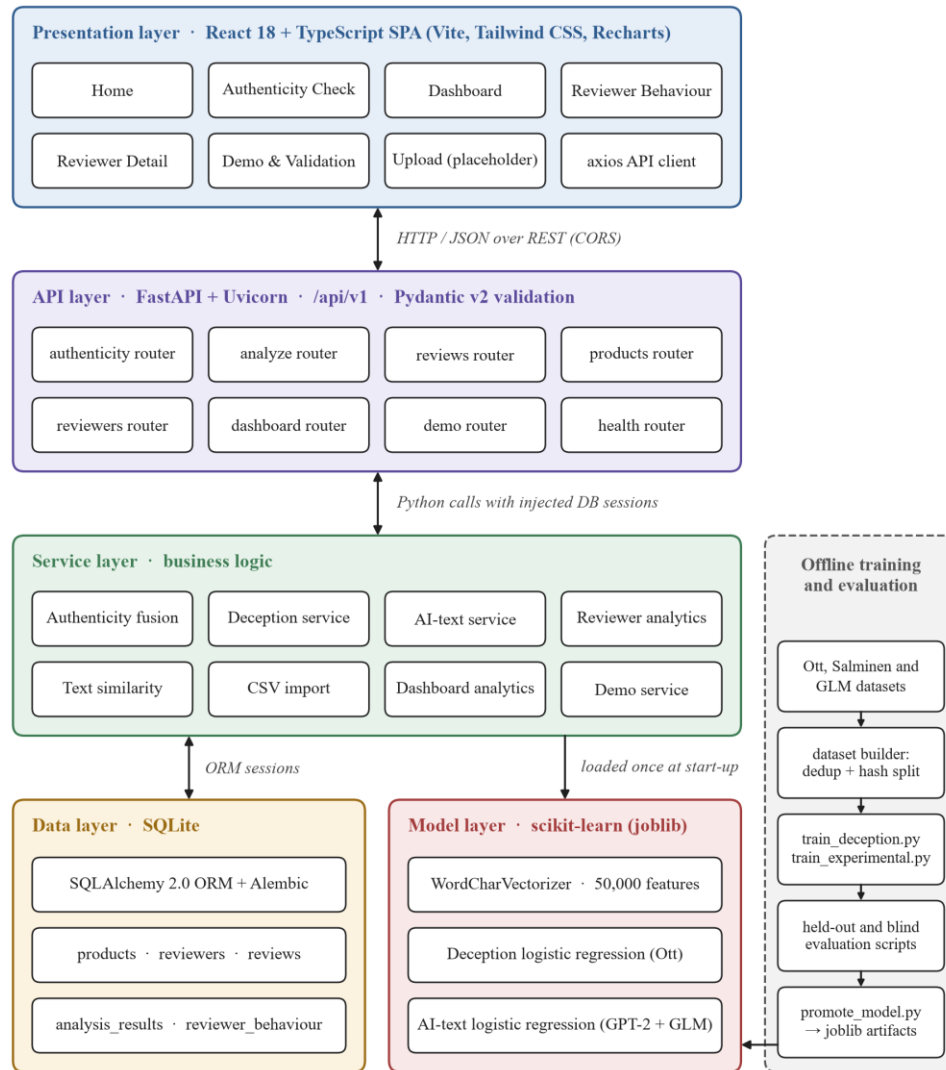


Figure 3.1. System architecture of the review authenticity platform

The **presentation layer** is written in TypeScript with React 18 [31], built with Vite and styled with Tailwind CSS. It provides the Home, Authenticity Check, Dashboard, Reviewer Behaviour, Reviewer Detail and Demo & Validation pages and a placeholder page for review upload, which is currently performed through the API; all HTTP calls pass through one axios client and the charts are drawn with Recharts. The **API layer** is a FastAPI application served by Uvicorn [32]. Eight routers expose 28 routes, request and response bodies are validated with Pydantic models, so malformed input such as an empty review is rejected with HTTP 422 before any model runs, and an interactive OpenAPI description is generated automatically.

The **service layer** contains the authenticity fusion service, the two model-serving services, reviewer analytics and text similarity, CSV import, dashboard aggregation and the demonstration service, with a database session injected into each request. The **model layer** holds the two trained scikit-learn pipelines [33], loaded once at start-up so that a request only performs vectorisation and a sparse matrix product. The **data layer** uses SQLAlchemy 2.0 with Alembic migrations over SQLite [34]; because all access goes through the ORM, the same code can run against PostgreSQL. The offline pipeline builds the datasets, trains and evaluates the models and promotes approved artifacts into the model layer.

3.2 SYSTEM REQUIREMENTS

3.2.1 SOFTWARE REQUIREMENTS

Table 3.1 lists the software used to build and run the system, with the versions installed in the verified development environment.

Table 3.1. Software requirements of the proposed system

Software component	Version	Purpose
Python	3.13.6	Back-end language, model training and evaluation
FastAPI, Uvicorn	0.115.6, 0.34.0	REST API framework and the ASGI server that runs it
Pydantic, pydantic-settings	2.10.4, 2.7.1	Request validation and environment-based configuration

Software component	Version	Purpose
SQLAlchemy, Alembic	2.0.36, 1.14.0	Object-relational mapping and schema migrations
SQLite	3.50.4	Relational database engine
scikit-learn	1.9.0	TF-IDF vectorisers, logistic regression and evaluation metrics
pandas, NumPy, SciPy	2.3.3, 2.5.1, 1.18.0	Dataset handling, sparse matrices and statistical tests
joblib	1.5.3	Serialisation of the trained models
pytest, httpx	8.3.4, 0.28.1	Automated back-end tests through the API test client
Node.js, npm	22.18.0, 10.9.3	Front-end runtime and package management
React, React Router	18.3, 6.30	Single-page user interface and client-side routing
TypeScript, Vite	7.0, 8.2	Typed front-end code and build tool
Tailwind CSS, Recharts, axios	3.4, 2.15, 1.19	Styling, dashboard charts and HTTP client
Git	2.54	Version control

3.2.2 HARDWARE REQUIREMENTS

The system was developed, trained and evaluated on the laptop described in Table 3.2. The deployed models are small: each vectoriser occupies about 1.7 MB and each classifier about 0.4 MB on disk, both models load in 0.36 seconds and scoring one review takes about 2.3 milliseconds. Neither training nor inference uses the graphics processor, so a dedicated GPU is not required.

Table 3.2. Hardware configuration used for development and evaluation

Component	Specification
Processor	Intel Core i5-13450HX, 10 cores and 16 threads
Memory	15.7 GB RAM
Storage	WD_BLACK SN770 500 GB and Samsung 512 GB solid-state drives
Graphics	NVIDIA GeForce RTX 4050 Laptop GPU and Intel UHD Graphics (not used by the system)
Operating system	Windows 11 Home Single Language, build 26200

3.2.3 DATASET DESCRIPTION

Three public datasets are used for training and one private set is used only for blind evaluation, as summarised in Table 3.3 and Figure 3.2.

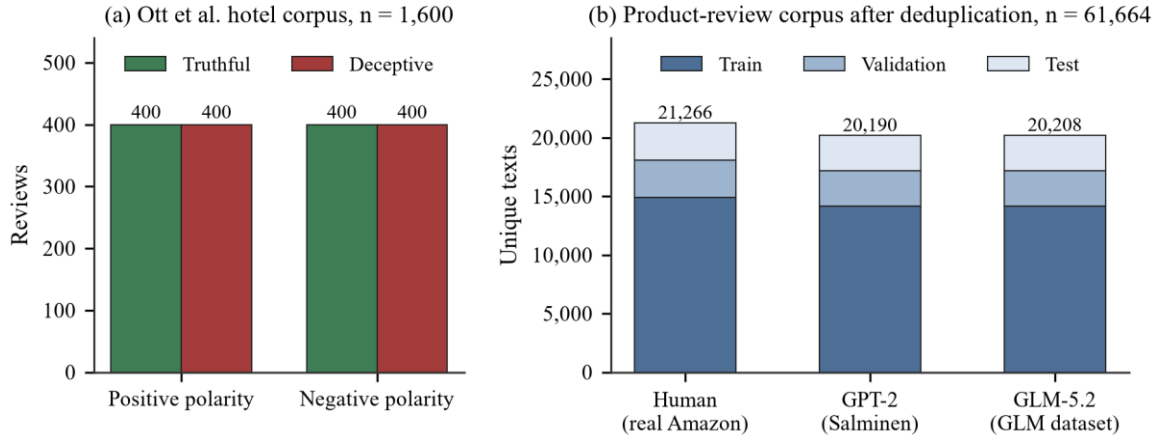
Table 3.3. Datasets used in the system

Dataset	Used for	Size and labels	Domain
Ott et al. Deceptive Opinion Spam Corpus v1.4 [2], [10]	Deceptive-writing model	1,600 reviews of 20 hotels: 800 truthful and 800 deceptive, written by paid crowdworkers	Hotels (Chicago)
Salminen et al. fake reviews dataset [12]	AI-text model	40,432 reviews: 20,216 real (OR) and 20,216 generated by GPT-2 (CG)	Amazon products
Flowerly/modern-fake-reviews-glm, CC-BY-4.0 [13]	AI-text model	40,416 reviews: 20,208 real and 20,208 generated by GLM-5.2	Amazon products
ChatGPT held-out set, supplied by the project team	Blind evaluation only	25 ChatGPT-written reviews, flagged never to be used for training	Products

The Salminen and GLM datasets share most of their human reviews: 94.8 percent of the human half of the GLM dataset is identical to reviews in the Salminen dataset. The two were therefore merged and deduplicated using a SHA-256 hash of the whitespace- and case-normalised text, which reduced 80,848 rows to 61,664 unique texts, and the corpus was split 70:15:15 over hash groups, stratified by source, so that no text appears in more than one split (Table 3.4). The product reviews cover ten Amazon categories with 5,490 to 7,268 texts each, and about 61 percent carry a five-star rating. Their median length is 226 characters for genuine, 174 for GPT-2 and 239 for GLM-5.2 reviews, whereas the Ott hotel reviews are much longer, with a median of 698 characters. The 25 ChatGPT reviews, with a median length of 190 characters, are stored separately with a flag that forbids their use in training, and an automated test fails if they could enter a training pipeline.

Table 3.4. Hash-grouped splits of the product-review corpus

Split	Human	GPT-2	GLM-5.2	Total
Train	14,886	14,133	14,145	43,164
Validation	3,189	3,028	3,031	9,248
Test	3,191	3,029	3,032	9,252
Total	21,266	20,190	20,208	61,664

**Figure 3.2. Composition of the training datasets**

3.2.4 DEPLOYMENT AND SCALING

In the verified configuration the back end runs under Uvicorn on port 8000 and the front end on the Vite development server on port 5173, with cross-origin requests allowed through a configurable CORS allow-list and settings such as the database URL read from environment variables. Docker files for both parts are included, but containerised deployment was not part of the verified workflow. The design can scale horizontally because the API keeps no session state outside the database, inference is a sparse matrix product with a batch throughput of about 1,700 reviews per second on the development machine, and the ORM allows SQLite to be replaced by PostgreSQL without code changes. Before public deployment, authentication, authorisation and rate limiting must be added, and dashboard aggregation, which currently loads analysis records into application memory, must move into the database.

3.3 SYSTEM DESIGN

3.3.1 USE CASE DIAGRAM

Figure 3.3 shows the use cases of the platform. An analyst or moderator checks the authenticity of individual reviews, reads the explanation and signal breakdown, monitors the dashboard, browses reviewer risk profiles and runs the demonstration cases. A data administrator imports review data from CSV files through the API, classifies the stored reviews and recomputes the reviewer-behaviour profiles. Because authentication is not implemented, the two actors describe responsibilities rather than enforced roles.

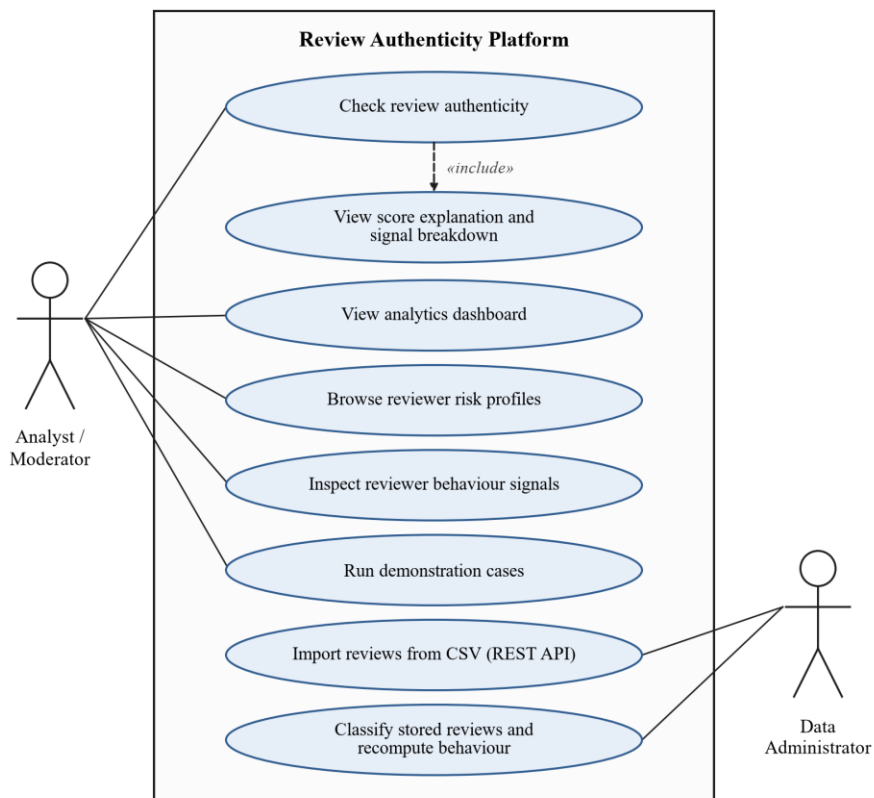


Figure 3.3. Use case diagram of the platform

3.3.2 ACTIVITY DIAGRAM

Figure 3.4 shows the activity flow of an authenticity check, which follows the control flow of the analysis endpoint. After validation the review is scored by both text models. If the request refers to a stored review, that review must exist and the behaviour profile of its reviewer is loaded. The request fails with HTTP 503 only when neither

model is available; otherwise the available signals are fused, the result is stored and the explanation is returned.

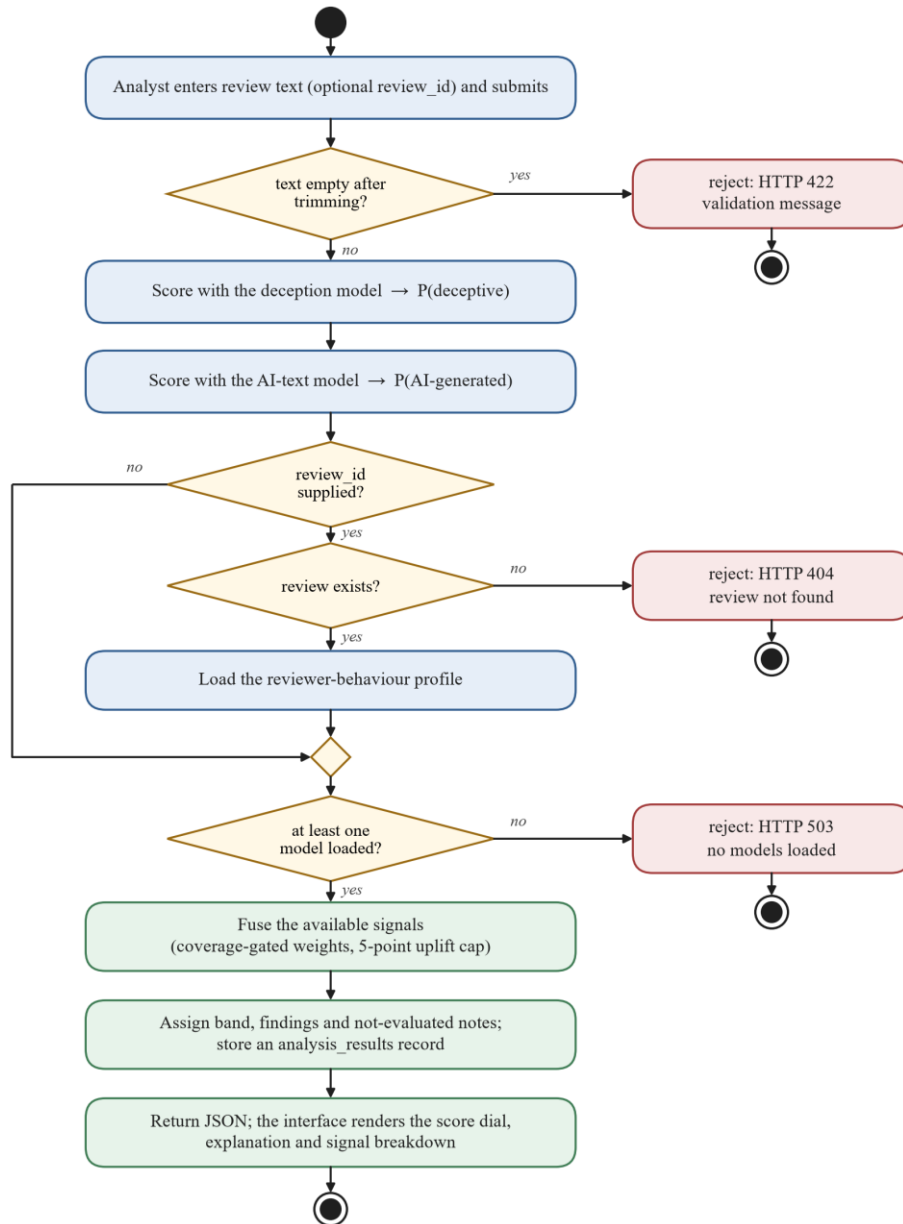


Figure 3.4. Activity diagram for checking the authenticity of a review

Review data enters the system through CSV upload. A file must be UTF-8 text with a header row containing product_name, reviewer_name, rating and review_body, otherwise it is rejected with HTTP 400; rows that fail validation are reported with their line numbers while the valid rows are imported. The stored reviews are then classified and the behaviour profile of every reviewer is recomputed, as described in Section 4.2.4.

3.3.3 DATA FLOW DIAGRAM

The context-level data flow diagram in Figure 3.5 treats the platform as a single process that exchanges data with the analyst, the data administrator and the labelled research datasets used for offline training and evaluation.

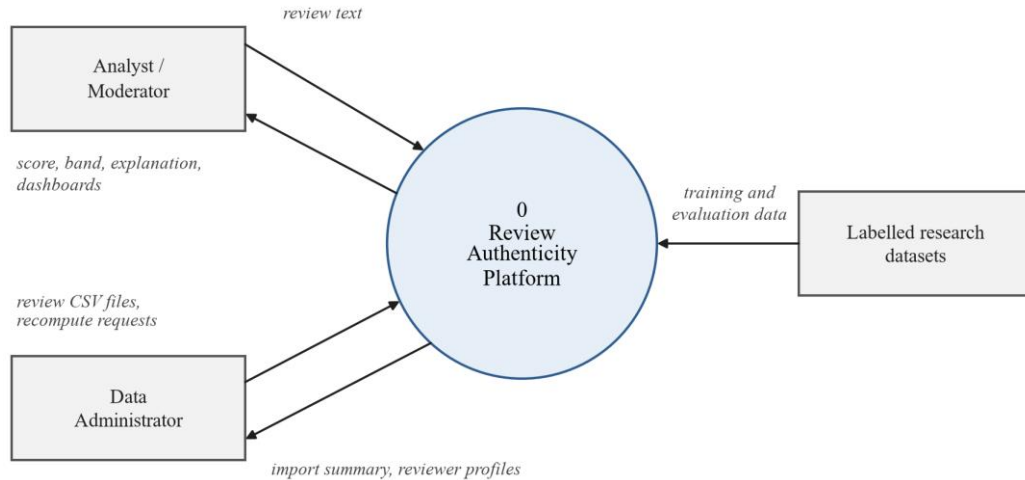


Figure 3.5. Level 0 data flow diagram

Figure 3.6 decomposes the platform into six processes and four data stores. Imported reviews (process 1.0) are kept in store D1 and, together with the stored predictions in D3, feed behaviour analysis (4.0), whose profiles are kept in D4. Review text is scored by the deception (2.0) and AI-text (3.0) processes using the model artifacts in D2, and the fusion process (5.0) combines their probabilities with the behaviour risk of the reviewer, stores an analysis record in D3 and returns the explanation, while dashboard analytics (6.0) aggregate the stored records.

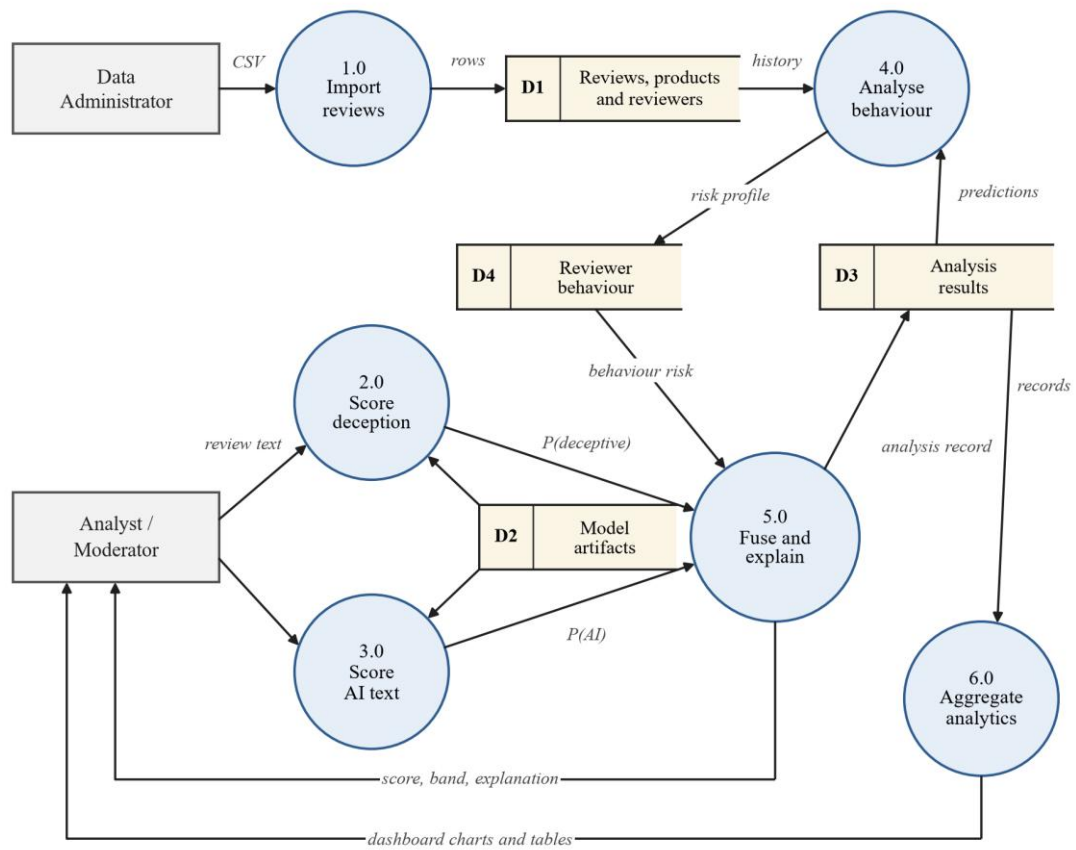


Figure 3.6. Level 1 data flow diagram

3.3.4 SEQUENCE DIAGRAM

Figure 3.7 details the messages exchanged during one authenticity request, from the action of the user in the browser to the rendered result. The optional fragment applies only when the request carries the identifier of a stored review.

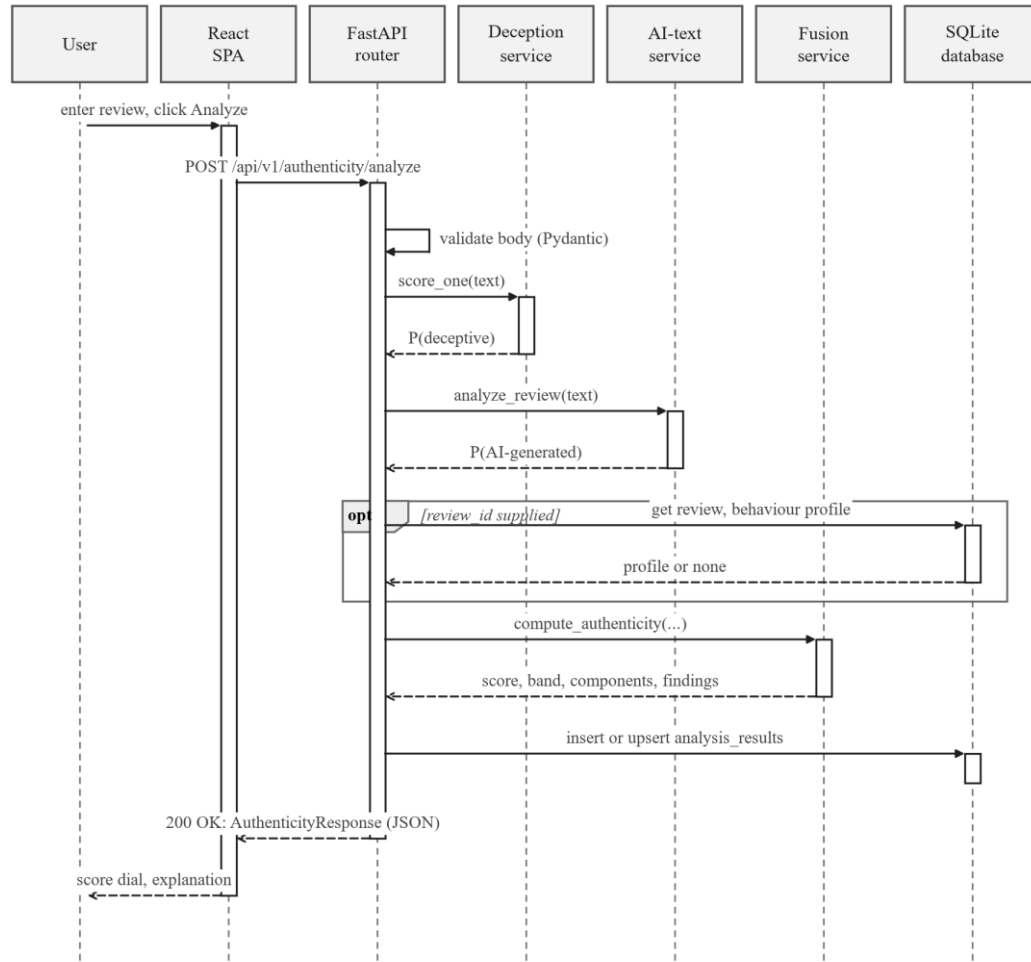


Figure 3.7. Sequence diagram of an authenticity analysis request

3.3.5 DATABASE DESIGN

The relational schema is shown in Figure 3.8. A product and a reviewer each have many reviews; a review has at most one linked analysis result and a reviewer at most one behaviour profile. The `review_id` of an analysis result is nullable so that ad-hoc analyses of pasted text can be stored. Ratings are constrained to the range 1–5, product codes and external reviewer identifiers are unique, foreign keys cascade on delete with enforcement switched on for every SQLite connection, and the indexed hash of each review body is computed automatically so that exact duplicates can be counted in linear time.

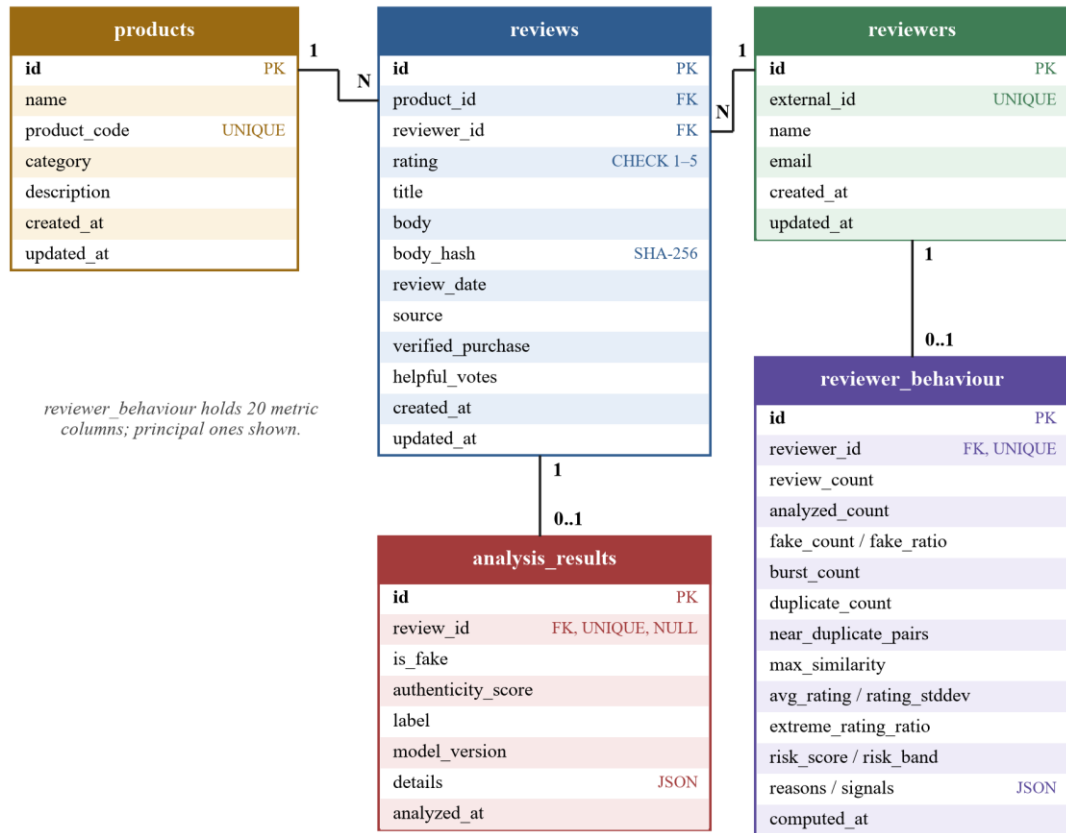


Figure 3.8. Entity–relationship diagram of the database

CHAPTER 4

PROJECT DESCRIPTION

4.1 METHODOLOGY

The methodology has two parallel model-development tracks, one for each text signal, followed by integration with the behaviour rules and the fusion procedure, as summarised in Figure 4.1.

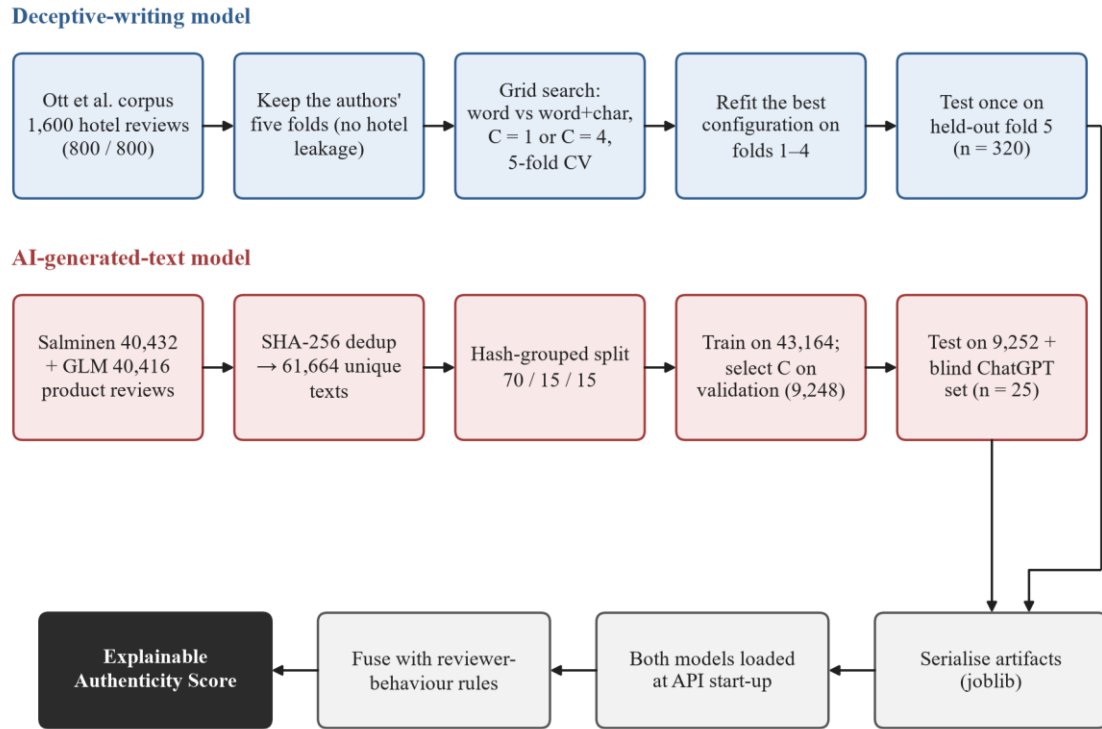


Figure 4.1. Model development and evaluation methodology

4.1.1 DATA PREPARATION AND LEAKAGE CONTROL

Data leakage, in which information from the evaluation data reaches a model during training, is a common cause of inflated results. Three controls were applied: the published five-fold assignment of the Ott corpus was kept, so that reviews of the same hotel are not scattered across training and test folds; every product review was keyed by the SHA-256 hash of its normalised text and splitting was performed over hash groups,

with assertions that no hash crosses a split boundary; and the dataset builder never loads the ChatGPT reviews. Table 4.1 lists the checks performed.

Table 4.1. Leakage checks performed

Check	Result
Duplicate texts within the Salminen dataset	25 groups, 52 rows (0.13%)
Original Salminen test rows also present in its training split	10 of 8,087 (0.12%)
Texts with conflicting labels	0
GLM-5.2 generated reviews overlapping Salminen GPT-2 reviews	0%
GLM human reviews identical to Salminen human reviews	94.8%, removed by deduplication before splitting
Hashes appearing in more than one split	0 (asserted in code)
ChatGPT held-out reviews in any training split	0
Ott fold-5 reviews used to fit the final deception model	0

During validation one leakage defect was found and corrected: an early evaluation sampled product reviews from the original Salminen split (random seed 42) although the AI-text model had been trained on the hash-grouped split (seed 7501), so 74 percent of those supposedly unseen reviews had been used in training. All results in Chapter 5 use samples drawn from the model's own test split.

4.1.2 FEATURE EXTRACTION

Both text models share the feature extractor WordCharVectorizer, shown with its parameters in Figure 4.2. It fits a word TF-IDF block of unigrams and bigrams (20,000 features, lowercased) and a character TF-IDF block of 3- to 5-grams within word boundaries (30,000 features, case preserved) and concatenates their sparse outputs. No cleaning, stop-word removal or stemming is applied, so capitalisation, punctuation and spacing remain available as evidence. Term frequency is dampened logarithmically (Equation 4.1), inverse document frequency lowers the weight of terms that occur in many of the n training documents (Equation 4.2), and each document vector is normalised to unit length (Equation 4.3) [35].

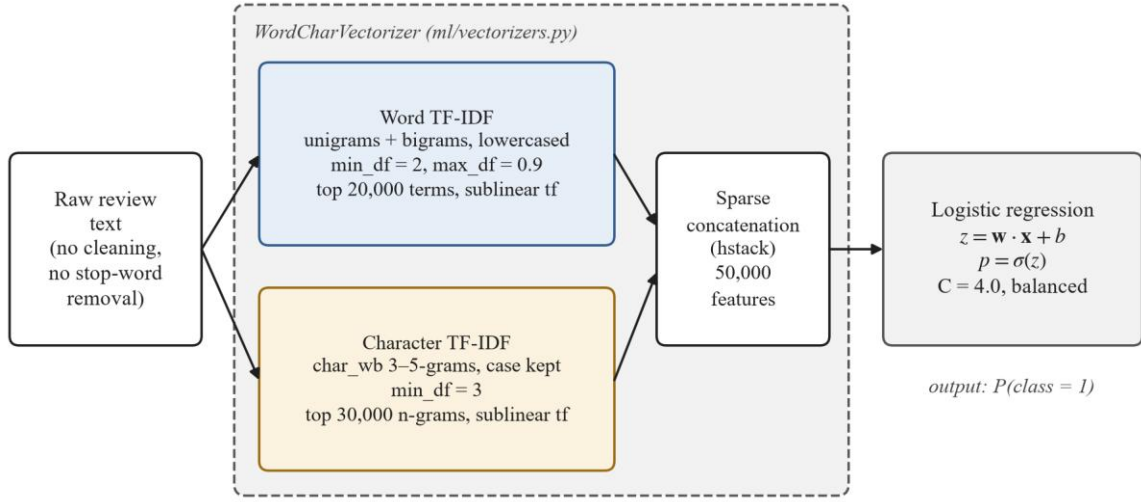


Figure 4.2. Word and character TF-IDF feature extraction

$$\text{tf}(t, d) = 1 + \ln \text{count}(t, d) \quad (4.1)$$

$$\text{idf}(t) = \ln \frac{1 + n}{1 + \text{df}(t)} + 1 \quad (4.2)$$

$$v_{t,d} = \text{tf}(t, d) \cdot \text{idf}(t), \quad \hat{\mathbf{v}}_d = \frac{\mathbf{v}_d}{\|\mathbf{v}_d\|_2} \quad (4.3)$$

The word block captures vocabulary and short phrases such as “my stay” or “and the”. The case-preserving character block captures typographic habits that a word model cannot see, such as a missing space after a full stop, repeated ellipses or sentence capitalisation; as Chapter 5 shows, such patterns are among the strongest cues of the AI-text model.

4.1.3 CLASSIFICATION MODEL

Both signals use logistic regression. For a feature vector $\mathbf{x}$ the model computes a linear score z and maps it to a probability with the sigmoid function (Equation 4.4); a review is assigned to the positive class, deceptive or AI-generated, when the probability is at least 0.5, and no other threshold is used in production. The weights minimise the L2-regularised logistic loss of Equation 4.5, in which the label of review i is written as $+1$ or -1 , C sets the strength of regularisation and c is the weight of the class of the review. Balanced class weights (Equation 4.6, with n samples, K classes and n_k samples in class

k) matter for the AI-text corpus, which contains roughly two AI-generated texts for every human one.

$$z = \mathbf{w}^\top \mathbf{x} + b, \quad P(y = 1 \mid \mathbf{x}) = \sigma(z) = \frac{1}{1 + e^{-z}} \quad (4.4)$$

$$\min_{\mathbf{w}, b} \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{i=1}^n c_{y_i} \ln(1 + e^{-\tilde{y}_i z_i}) \quad (4.5)$$

$$c_k = \frac{n}{K \cdot n_k} \quad (4.6)$$

Logistic regression suits this task because sparse, high-dimensional TF-IDF data favour linear models, the convex training problem gives reproducible solutions, the output probabilities can be fused directly, and every coefficient can be inspected to show which features drove a decision. Table 4.2 lists the settings of the two production models.

Table 4.2. Hyperparameters and training of the two text models

Setting	Deceptive-writing model	AI-text model
Solver and class weight	lbfgs, balanced	lbfgs, balanced
Regularisation C and maximum iterations	4.0 and 3,000	4.0 and 3,000
Random state	7502	7501
Selection	word vs word + character features and C = 1 or 4, by 5-fold CV accuracy	C chosen by F1-score on the validation split
Training data	Ott folds 1–4 (1,280 reviews)	43,164 deduplicated texts
Artifacts	deception_vectorizer.joblib, deception_model.joblib	vectorizer.joblib, model.joblib

4.1.4 AUTHENTICITY SCORE FUSION

The fusion procedure of Figure 4.3 combines the risks of the available signals. Each text signal contributes its probability as a risk r between 0 and 1 and the behaviour signal its risk score divided by 100. With A the set of available signals, the combined risk is the weighted mean in Equation 4.7, with weights of 0.45 for deception, 0.25 for AI-generated text and 0.30 for behaviour, so a text-only assessment is still a proper weighted mean of the two text signals.

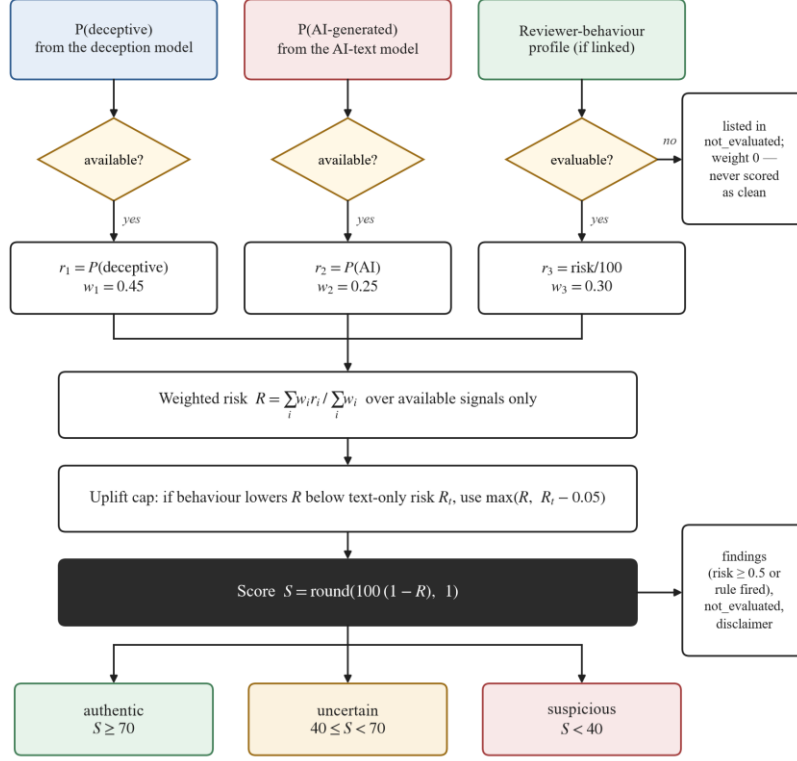


Figure 4.3. Authenticity Score fusion procedure

$$R = \frac{\sum_{j \in A} w_j r_j}{\sum_{j \in A} w_j} \quad (4.7)$$

Because a clean history is only weak evidence that a particular review is genuine, whereas bursts or duplicates are strong evidence against it, an asymmetric cap limits any reduction of risk by behaviour to 0.05 below the text-only risk R_T (Equation 4.8). The score follows from Equation 4.9 and is mapped to the bands of Table 4.3.

$$R^* = \max(R, R_T - 0.05) \text{ if } R < R_T, \text{ otherwise } R^* = R \quad (4.8)$$

$$S = \text{round}(100(1 - R^*), 1) \quad (4.9)$$

Table 4.3. Signal weights and score bands

Item	Value	Meaning
Deception weight	0.45	On-target signal for paid human deception
AI-text weight	0.25	Partial signal; 64% of blind ChatGPT reviews detected
Behaviour weight	0.30	Used only when the reviewer profile is evaluable
Behaviour uplift cap	0.05 (5 points)	A clean history cannot hide flagged text
Authentic band	$S \geq 70$	No strong risk signal in the checks that ran
Uncertain band	$40 \leq S < 70$	Some risk signals; worth a human look
Suspicious band	$S < 40$	Multiple or strong signals; manual review recommended

A behaviour profile is evaluable only if a rule fired or the reviewer has at least three analysed reviews; an unavailable signal is listed among the checks not run and is never scored as clean. Findings are emitted only when a text probability is at least 0.5 or a behaviour rule fired. The weights and band limits are engineering choices informed by validation rather than calibrated values, because no labelled product-review fraud dataset is publicly available.

For example, the review “this is a really good product , but it is a little expensive and has less color options.” receives $P(\text{deceptive}) = 0.187$ and $P(\text{AI-generated}) = 0.040$ and has no reviewer history, so $R = (0.45 \times 0.187 + 0.25 \times 0.040) / 0.70 = 0.134$ and $S = 100 \times (1 - 0.134) = 86.6$, which lies in the authentic band.

4.1.5 EVALUATION METRICS

With fake reviews as the positive class and TP, TN, FP and FN the numbers of true positives, true negatives, false positives and false negatives, **accuracy** and **precision** are given by Equation 4.10, **recall** (the detection rate) and the **F1-score** by Equation 4.11, and the **false-positive rate**, the share of genuine reviews wrongly flagged, by Equation 4.12. **ROC-AUC** is the area under the curve of true-positive rate against false-positive rate over all thresholds; it equals the probability that a random fake review receives a higher risk than a random genuine one, so 0.5 is chance and 1.0 is perfect separation.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}, \quad \text{Precision} = \frac{TP}{TP + FP} \quad (4.10)$$

$$\text{Recall} = \frac{TP}{TP + FN}, \quad F_1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4.11)$$

$$\text{FPR} = \frac{FP}{FP + TN} \quad (4.12)$$

Fisher's exact test [36] on detection counts and the Wilcoxon signed-rank test [37] on paired probabilities were used to check whether the improvement on ChatGPT reviews could be due to chance, and detection rates measured on small sets are reported with 95 percent binomial confidence intervals.

4.2 MODULES

The system is organised into the six modules described below.

4.2.1 MODULE 1: REVIEW DATA INGESTION

This module imports review data and maintains the relational store. The upload endpoint accepts a UTF-8 CSV file with the required columns `product_name`, `reviewer_name`, `rating` and `review_body` and optional columns such as `review_date`, `category` and `helpful_votes`. Products and reviewers are created or reused, invalid rows are reported individually instead of aborting the file, and the hash of each review body is maintained by a database event so that duplicates can be counted efficiently.

4.2.2 MODULE 2: DECEPTIVE REVIEW DETECTION

The deception service loads its vectoriser and model at start-up and returns $P(\text{deceptive})$ for raw text. The model targets reviews written by people paid to invent them. Because it was trained and validated on hotel reviews, it acts on product reviews as a supporting signal whose transfer to that domain has not been measured, a caveat recorded in its model report.

4.2.3 MODULE 3: AI-TEXT DETECTION

The machine-learning service returns the predicted label, the confidence and the class probabilities for each text. Training on GPT-2 and GLM-5.2 reviews together with real Amazon reviews encourages the model to learn cues shared by different generators

rather than the quirks of one. The earlier GPT-2-only model is kept on disk as a baseline but is not loaded, and a model card endpoint states what the deployed classifier can and cannot detect.

4.2.4 MODULE 4: BEHAVIOUR ANALYSIS

The reviewer analytics service builds a behaviour profile for every reviewer: review volume, the fake-review ratio among analysed reviews, posting velocity, burst days with five or more reviews, exact duplicates found through the body hash, near-duplicates found through the cosine similarity of character 3-gram TF-IDF vectors (Equation 4.13) and rating statistics. Similarity is computed only within the reviews of one reviewer, texts under 20 characters are ignored and reviewers with more than 300 reviews are skipped and reported. The six weighted rules of Table 4.4 convert these measurements into a risk score capped at 100 (Equation 4.14), banded as low below 40, medium from 40 and high from 70. A rule that cannot be evaluated produces a data-quality note rather than a silent zero, and because the fake-review ratio comes from stored AI-text predictions, that rule is not fully independent of the AI-text signal.

$$\text{sim}(\mathbf{a}, \mathbf{b}) = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|} \quad (4.13)$$

$$\text{Risk} = \min \left(100, \sum_{k=1}^6 s_k \right) \quad (4.14)$$

Table 4.4. Reviewer-behaviour rules

Rule	Trigger	Contribution s
High fake-review ratio	at least 3 analysed reviews and fake ratio ≥ 0.5	$30 \times (0.5 + 0.5 \times (\text{ratio} - 0.5) / 0.5)$, at most 30
Duplicate reviews	at least one exact duplicate	$\min(25, 10 + 5 \times \text{duplicates})$
Near-duplicate reviews	at least one pair with cosine similarity ≥ 0.85	$\min(20, 8 + 4 \times \text{pairs})$
Posting burst	at least one day with 5 or more reviews	$\min(15, 8 + 3.5 \times \text{burst days})$
High posting velocity	at least 3 reviews per active day and at least 2 reviews	$\min(10, 5 + \text{reviews per active day})$
Extreme ratings	at least 5 reviews and at least 90% one- or five-star ratings	10

4.2.5 MODULE 5: AUTHENTICITY SCORING

This module implements the fusion of Section 4.1.4 behind POST /api/v1/authenticity/analyze. The request carries the review text and optionally a stored review identifier; the response returns the score, band, mode, per-signal components, findings, checks not evaluated and a disclaimer that the score is a triage aid rather than proof. Every assessment is persisted, so the dashboard reflects real use.

4.2.6 MODULE 6: WEB APPLICATION

The React application offers the Home, Authenticity Check, Dashboard, Reviewer Behaviour, Reviewer Detail and Demo & Validation pages. The principal REST endpoints are listed in Table 4.5; the remaining routes cover product and review management, the model card, reviewer history, behaviour recomputation, a validation report and health checks, for 28 routes in total. The demonstration mode scores eight documented cases through the live endpoint, and none of its data is used for training.

Table 4.5. Principal REST API endpoints

Method	Endpoint	Purpose
POST	/api/v1/authenticity/analyze	Compute the Review Authenticity Score
POST	/api/v1/reviews/upload	Import reviews from a CSV file
POST	/api/v1/reviews/analyze-stored	Classify stored reviews and link the results
GET	/api/v1/reviewers, /{id}, /{id}/behaviour	Reviewer list, profile and behaviour signals
GET	/api/v1/dashboard/summary, /charts, /analyses	Dashboard statistics, chart series and table
POST	/api/v1/analyze, /api/v1/analyze/batch	AI-text prediction for one text or up to 100 texts
GET, POST	/api/v1/demo/cases, /api/v1/demo/seed	Demonstration cases and scenario seeding

CHAPTER 5

IMPLEMENTATION AND RESULT DISCUSSION

5.1 IMPLEMENTATION RESULTS

This section presents the working platform and the measured performance of each component. All results were produced from the production model artifacts by read-only evaluation scripts, and no model was trained or tuned on the evaluation data.

5.1.1 WEB PLATFORM

The Authenticity Check page is the primary workflow. Figure 5.1 shows the result for a heavily promotional review. The deceptive-writing model gives it a probability of 0.64 and the AI-text model 0.90, so both findings are listed. Because the pasted text is not linked to a reviewer, the behaviour check is reported as not evaluated, and the score of 26.7, displayed as 27, comes from the two text signals alone. For the worked example of Section 4.1.4 the same page shows a score of 86.6 in the authentic band with no findings. In both cases the signal breakdown lists the risk and weight of every signal, and the page states that the score is a triage aid rather than proof.

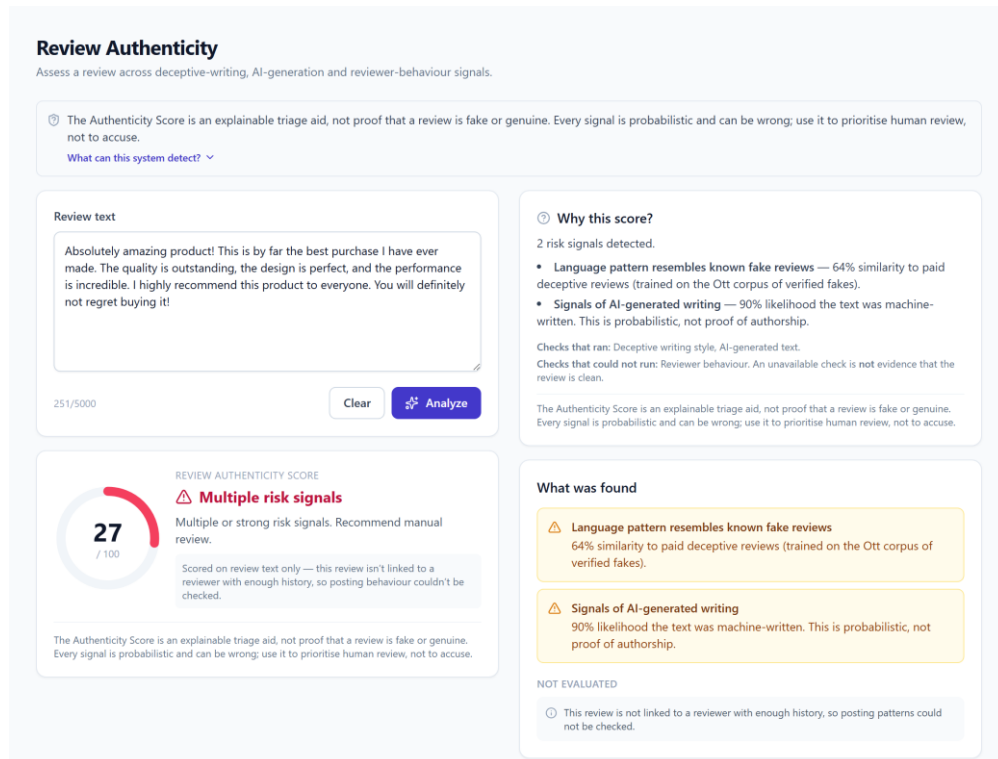


Figure 5.1. Authenticity Check result for a promotional, AI-style review

The dashboard in Figure 5.2 summarises the stored assessments with totals, the split between fake and genuine assessments, the confidence distribution and the number of analyses per day, followed by a searchable table. An assessment counts as fake only when it falls in the suspicious band, and the values reflect development and testing use rather than a labelled benchmark.

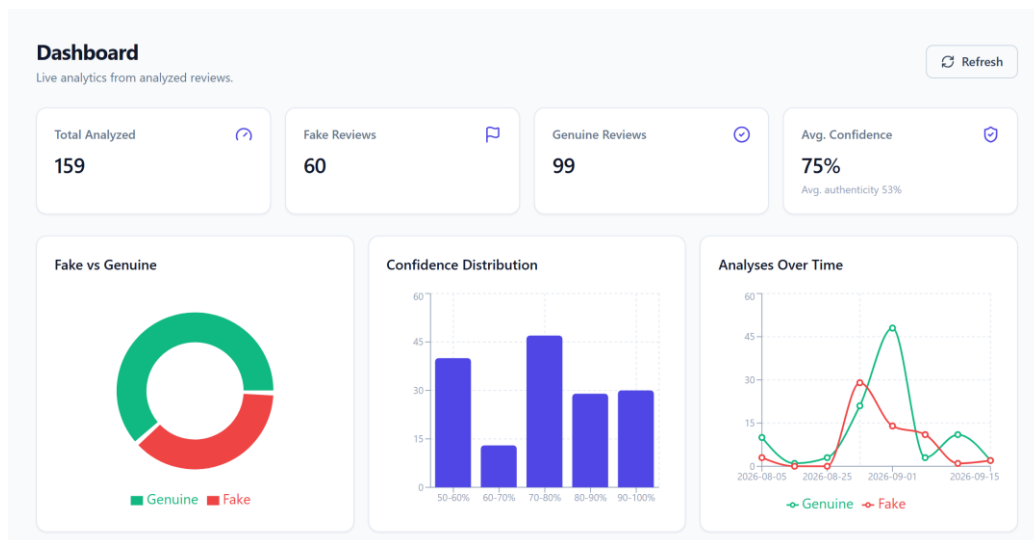


Figure 5.2. Analytics dashboard

The Reviewer Behaviour pages list reviewers by risk and explain each profile. Figure 5.3 shows the seeded duplicate-review scenario, whose risk of 67.5, displayed as 68, comes from a fake-review ratio of 75 percent (+22.5), three duplicate reviews (+25) and six near-duplicate pairs (+20); the rating rule is listed as not evaluated because it needs at least five reviews. In the seeded burst scenario, six five-star reviews posted on one day trigger the fake-ratio (+20), burst (+11.5), velocity (+10) and extreme-rating (+10) rules, giving a risk of 51.5.

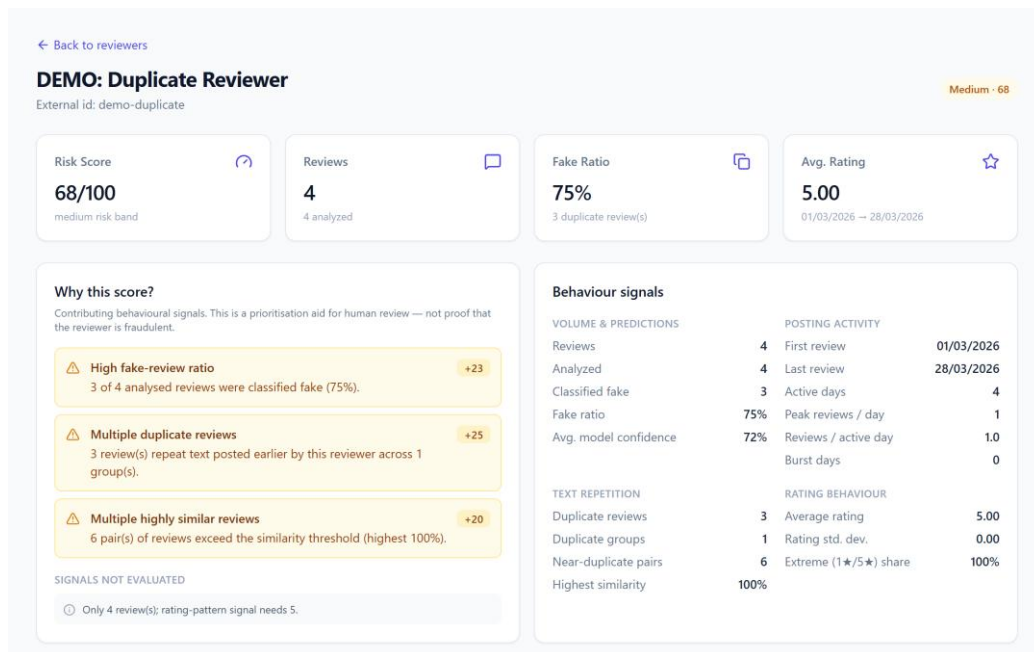


Figure 5.3. Profile of the duplicate-review demonstration reviewer

The Demo & Validation page runs eight documented cases through the live endpoint, with the scores listed in Table 5.1. The genuine product and hotel controls scored 88 and 75, while the ChatGPT and GPT-2 excerpts (52 and 59) and the human-written deceptive hotel review (42) fell in the uncertain band. The very short genuine review scored 58, an instance of the short-text weakness discussed in Section 5.2, and the two behaviour scenarios, scored together with their seeded reviewer histories, received 45 and 47.

Table 5.1. Scores of the demonstration cases returned by the live API

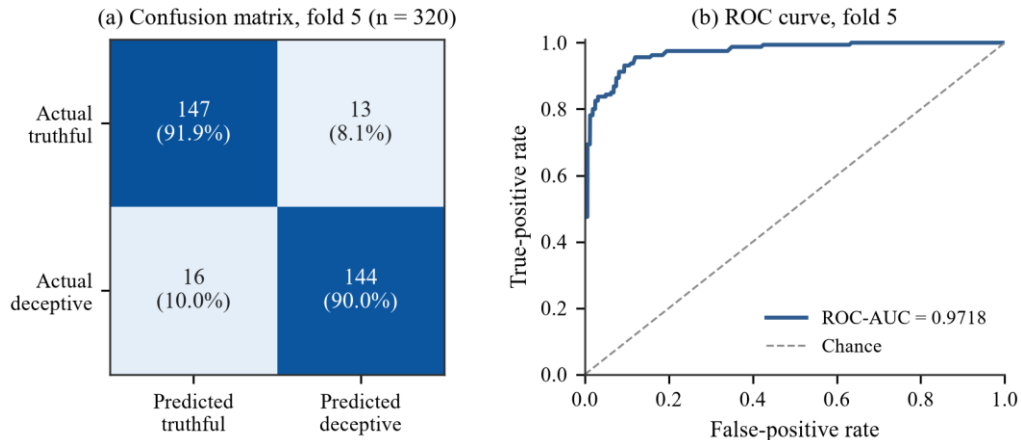
Demonstration case	Tests	Expected	Score	Band
Genuine product review	negative control	genuine	88	authentic
AI-generated product review (ChatGPT)	AI text	AI-generated	52	uncertain
AI-generated product review (GPT-2)	AI text	AI-generated	59	uncertain
Human-written deceptive review (hotel)	deception	human deceptive	42	uncertain
Genuine hotel review (control)	deception	genuine	75	authentic
Very short genuine review	short-text robustness	genuine	58	uncertain
Duplicate review / copied text	behaviour	suspicious behaviour	45	uncertain
Suspicious high-activity reviewer	behaviour	suspicious behaviour	47	uncertain

5.1.2 PERFORMANCE OF THE TEXT MODELS

For the deceptive-writing model, four configurations were compared by five-fold cross-validation on the published folds of the Ott corpus. Character n-grams improved accuracy over word n-grams alone, and word plus character n-grams with $C = 4$ was selected, with a mean accuracy of 89.1 percent and the smallest variation across folds. Refitted on folds 1–4 and evaluated on the 320 reviews of fold 5, the model reached an accuracy of 90.94 percent and a ROC-AUC of 0.9718 (Table 5.2, Figure 5.4), wrongly flagging 13 of 160 truthful reviews and missing 16 of 160 deceptive ones. Because fold 5 also served as one of the validation folds when choosing among the four configurations, the cross-validation mean of 89.1 percent is the more conservative estimate of in-domain accuracy.

Table 5.2. Classification metrics of the two text models

Metric	Deceptive-writing model (Ott fold 5, n = 320)	AI-text model (test split, n = 9,252)
Accuracy	0.9094	0.9422
Precision	0.9172	0.9661
Recall	0.9000	0.9449
F1-score	0.9085	0.9554
ROC-AUC	0.9718	0.9865
False-positive rate	0.0813	0.0630
False-negative rate	0.1000	0.0551
Confusion matrix [TN FP; FN TP]	[147 13; 16 144]	[2,990 201; 334 5,727]
Model selection	5-fold CV accuracy 0.891 ± 0.013	validation F1-score 0.9588

**Figure 5.4. Confusion matrix and ROC curve of the deceptive-writing model on fold 5**

On the 9,252-review test split the AI-text model reached an accuracy of 94.22 percent and a ROC-AUC of 0.9865 (Figure 5.5). Table 5.3 breaks the results down by population: the model detected 91.65 percent of GPT-2 and 97.33 percent of GLM-5.2 reviews while flagging 6.30 percent of genuine reviews, and it detected 16 of the 25 ChatGPT reviews, which come from a generator absent from training. Figure 5.6 shows why ChatGPT is harder: genuine reviews concentrate near zero and GPT-2 and GLM-5.2 reviews near one, while the ChatGPT reviews spread across the whole range.

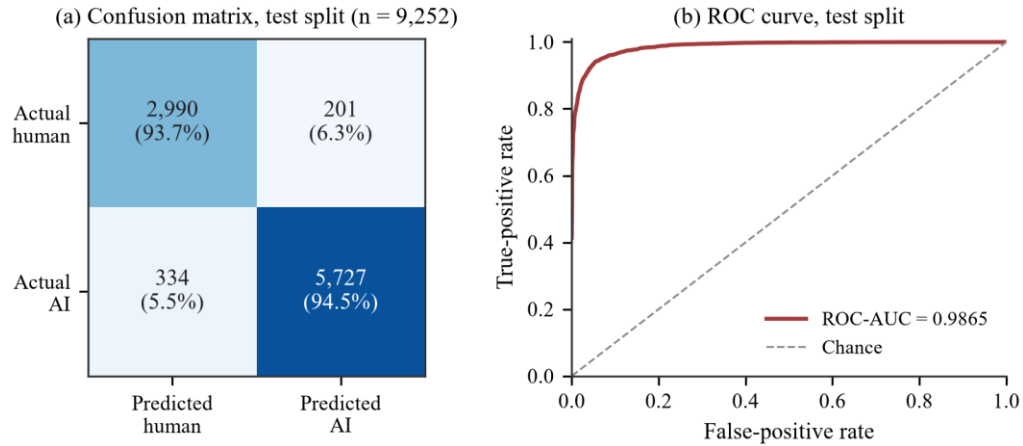


Figure 5.5. Confusion matrix and ROC curve of the AI-text model on the test split

Table 5.3. Detection by population for the AI-text model

Population	n	Result
GPT-2 reviews (test split)	3,029	91.65% detected
GLM-5.2 reviews (test split)	3,032	97.33% detected
Genuine human reviews (test split)	3,191	6.30% flagged (false positives)
ChatGPT reviews (blind, never used in training)	25	64.0% detected (16 of 25); 95% CI 42.5–82.0%
ChatGPT vs genuine human reviews	25 and 3,191	ROC-AUC 0.8652

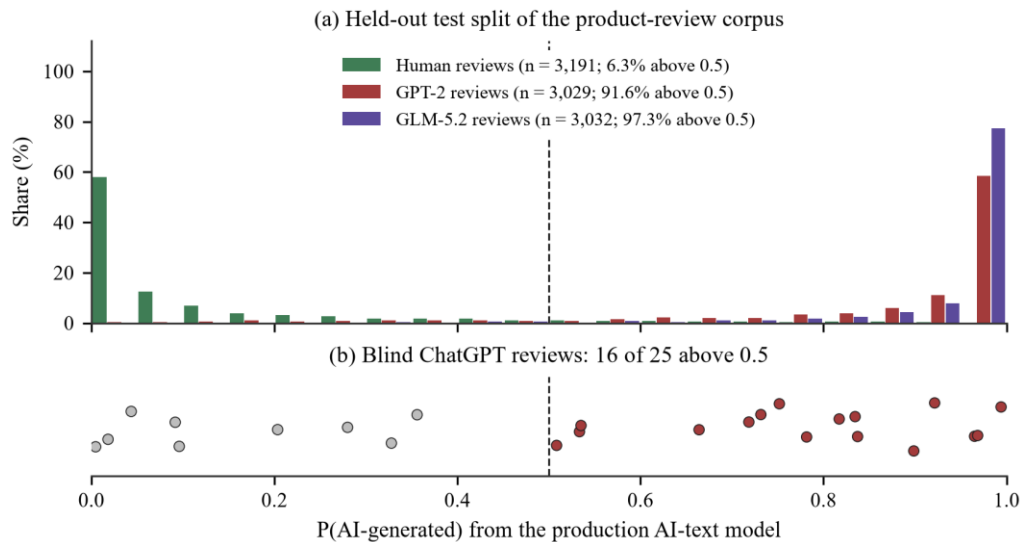


Figure 5.6. Distribution of P(AI-generated) by source, including the blind ChatGPT reviews

Table 5.4 shows the benefit of training on more than one generator. The earlier production model, trained on GPT-2 reviews with case and punctuation removed,

detected 2.67 percent of GLM-5.2 and 8 percent of ChatGPT reviews, and its ChatGPT-versus-human ROC-AUC of 0.505 was no better than chance. Character features without new training data (EXP-A) did not help, whereas adding GLM-5.2 data (EXP-B, now in production) raised these figures to 97.33 percent, 64 percent and 0.865. The improvement on ChatGPT from 2 to 16 of 25 reviews is statistically significant (Fisher's exact test, $p = 0.00007$; Wilcoxon signed-rank test on the paired probabilities, $p = 0.000015$). Because the baseline was trained on a different split of the Salminen data, its GPT-2 and human figures on this split would be optimistic and are not compared.

Table 5.4. Single-generator versus multi-generator training

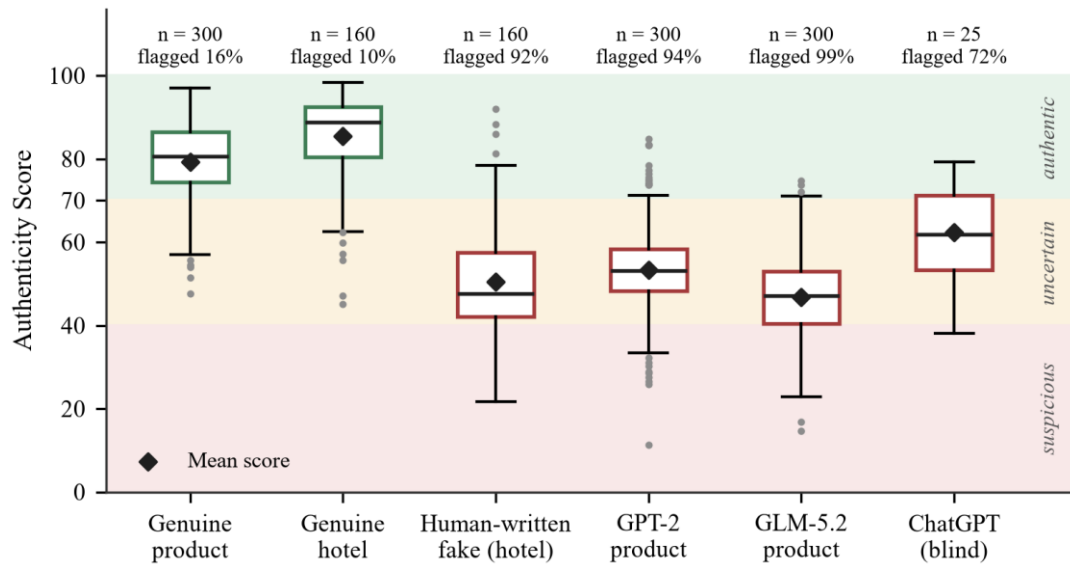
Model	Training data and features	GLM-5.2 detected	ChatGPT detected	ChatGPT vs human AUC	Test split AUC
Baseline	GPT-2 only; cleaned text, word TF-IDF	2.67%	8% (2/25)	0.5054	0.6730
EXP-A	GPT-2 only; raw text, word and character TF-IDF	0.66%	4% (1/25)	0.5975	0.6757
EXP-B (production)	GPT-2 and GLM-5.2; raw text, word and character TF-IDF	97.33%	64% (16/25)	0.8652	0.9865

5.1.3 COMBINED AUTHENTICITY SCORE

The fused score was evaluated in text-only mode on two leakage-free sets. Evaluation set E1 contains 300 random test-split reviews from each product population, the 320 reviews of Ott fold 5 and the 25 ChatGPT reviews (Table 5.5, Figure 5.7). Genuine populations score mostly in the authentic band and every fake population has a mean below 70, although most fake reviews fall in the uncertain band and the two distributions overlap near the 70 cut-off, as expected of a triage aid rather than a verdict.

Table 5.5. Authenticity Score by population on evaluation set E1

Population	n	Ground truth	Mean score	Flagged (not authentic)
Genuine product reviews (Amazon)	300	genuine	79.3	16%
Genuine hotel reviews (Ott truthful)	160	genuine	85.4	10%
Human-written fake reviews (Ott deceptive)	160	fake	50.5	92%
GPT-2 product reviews	300	fake	53.3	94%
GLM-5.2 product reviews	300	fake	46.8	99%
ChatGPT product reviews (blind)	25	fake	62.3	72%

**Figure 5.7. Authenticity Score distribution by population on evaluation set E1**

Evaluation set E2 pools 400 genuine product reviews, 300 short genuine product reviews under 200 characters, the 320 Ott fold-5 reviews and 300 GPT-2, 300 GLM-5.2 and 25 ChatGPT reviews, 1,645 in total, and counts any band other than authentic as a positive prediction. On E2 the fused score achieved an accuracy of 0.885, a precision of 0.831, a recall of 0.953, an F1-score of 0.888 and a ROC-AUC of 0.952, with a false-positive rate of 0.177 and a false-negative rate of 0.047; the short genuine reviews account for 68 of the 152 false positives. On the same set the deception signal alone reaches a ROC-AUC of only 0.635 and the AI-text signal alone 0.842, so neither signal is sufficient on its own.

5.1.4 BEHAVIOUR, TESTING AND PERFORMANCE

The effect of behaviour was measured on a review whose two text probabilities are both 0.20, which scores 80.0 without reviewer history. An evaluable history in which no rule fired raises the score only to 85.0 because of the uplift cap, whereas reviewer risks of 40, 80 and 100 lower it to 74.0, 62.0 and 56.0, moving the review from the authentic to the uncertain band in the last two cases. Behaviour is therefore the only signal that can expose a copied review whose text is itself genuine human writing.

The back end is covered by 175 automated pytest tests in 17 files, all of which pass, including integrity tests that fail the build if the ChatGPT held-out reviews or the demonstration cases could enter training or if reported metrics stop matching the stored artifacts, and the front end passes TypeScript type checking. Measured in-process on the development machine, loading both models takes 0.36 seconds, scoring one review takes 2.26 milliseconds at the median and 4.84 milliseconds at the 95th percentile, and a batch of 1,000 reviews takes 0.58 seconds, about 1,730 reviews per second.

5.2 DISCUSSION

Complementary signals. Figure 5.8 shows the share of each E1 population that each text signal places above 0.5. The deception signal catches 90 percent of the human-written fakes but only 4 to 10 percent of the machine-generated reviews, while the AI-text signal catches 92 percent of GPT-2, 96 percent of GLM-5.2 and 64 percent of ChatGPT reviews but only 2 percent of the human-written fakes, and the correlation between the two probabilities is -0.246 . The signals therefore cover different kinds of fake review, which justifies keeping them separate and fusing them.

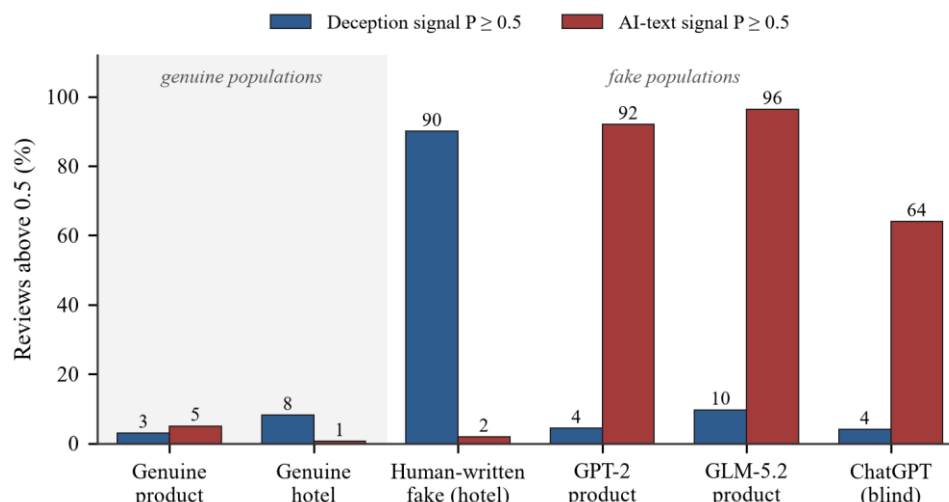


Figure 5.8. Share of reviews above 0.5 for each text signal, by population

What the models learned. The deception model rewards first-person and experiential framing such as “my” (+2.21), “my stay” (+1.59) and “vacation” (+1.22) and penalises concrete details such as “location” (−1.82) and “floor” (−1.58), in line with Ott et al. [2], but its strongest feature is “chicago” (+2.28), so part of what it learned is specific to the hotel corpus. The AI-text model relies on phrasing and typography: “and the” (+10.66), a full stop followed directly by “I” (+9.39) and “honestly” (+6.86) push towards AI-generated text, while “it” (−7.45), “to” (−7.31) and ellipses (−7.00) push towards human writing.

Short reviews. Review length is the largest source of false positives: 30 percent of genuine product reviews under 80 characters and 27 percent of those between 80 and 200 characters are flagged, against 16 percent between 200 and 600 characters and 6 percent above 1,000 characters. The mean AI-text probability of genuine reviews rises from 0.015 for the longest to 0.210 for the shortest, because a short text offers few features and a single typographic habit carries more weight.

Domain transfer and new generators. No public corpus of verified human-written fake product reviews exists, so the detection rate of the deception model on product reviews cannot be measured. Removing hotel vocabulary from its training data cost 1.6 points of in-domain accuracy but raised false positives on genuine product reviews from 3.7 to 18.3 percent, and a validation weight sweep found that for product reviews the AI-text signal alone ranks reviews slightly better (ROC-AUC 0.9877 against

0.9828), consistent with the cross-domain degradation reported in the literature. The ChatGPT result rests on only 25 reviews, and casually written AI text can evade the typography-sensitive AI-text model.

An evaluated improvement not yet deployed. A proposed scoring policy abstains on reviews shorter than 120 characters and counts the deception signal only when the text contains at least two hotel-domain marker words. Evaluated on E2 without changing production, it lowered the false-positive rate from 17.7 to 12.4 percent and raised precision from 0.831 to 0.888, the F1-score from 0.888 to 0.917 and the ROC-AUC from 0.952 to 0.961 on the 1,233 reviews it assessed, with recall almost unchanged. The cost is coverage: it declined to score 23.5 percent of genuine product, 54.3 percent of short genuine, 35.3 percent of GPT-2 and 16.3 percent of GLM-5.2 reviews, and because its metrics cover only the assessed reviews, it remains experimental until this trade-off is accepted.

Limitations. Table 5.6 summarises the known limitations and their status. Taken together, the results show that the system meets its objectives within measured and documented boundaries: it detects human-written deception in the domain on which it was validated, detects AI-generated reviews including a substantial share from an unseen generator, exposes suspicious reviewer behaviour with explicit reasons and produces an explainable score whose limitations are stated rather than hidden.

Table 5.6. Known limitations of the current system

Limitation	Effect	Status
Deception model trained on hotel reviews	Detection of human-written fake product reviews is unmeasured	Open; needs a labelled product corpus
Short genuine reviews	27–30% flagged below 200 characters	Abstention evaluated, not deployed
Blind ChatGPT set of 25 reviews	Wide confidence interval; other models untested	Open
Weights and bands not calibrated on product fraud data	Genuine and fake score distributions overlap near the 70 cut-off	Open
Fake-review ratio derived from AI-text predictions	Behaviour and AI-text signals not fully independent	Documented
No detection of coordinated multi-account campaigns	Groups of related accounts are not linked	Future work
No authentication or rate limiting	Not ready for public deployment	Future work

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 CONCLUSION

This project developed an AI-powered product review authenticity and fake review detection system that treats authenticity as a question with several kinds of evidence. Human-written deception and AI-generated text are detected by two separately trained text models, suspicious posting behaviour is detected by six explainable rules, and the three signals are fused into a Review Authenticity Score from 0 to 100 that shows the contribution of each signal, lists the findings behind every deduction and reports the checks that could not be run. The system is delivered as a complete platform: a FastAPI back end with a relational database and a documented REST API, and a React application with an authenticity checker, an analytics dashboard, reviewer-behaviour pages and a demonstration mode.

Every component was validated on data it had not been trained on, with explicit leakage controls. The deceptive-writing model achieved 90.9 percent accuracy and a ROC-AUC of 0.972 on held-out hotel reviews. The AI-text model detected 91.7 percent of GPT-2 and 97.3 percent of GLM-5.2 reviews and 64 percent of blind ChatGPT reviews, a statistically significant improvement over the single-generator baseline, which performed at chance on the unseen generator. On 1,645 evaluation reviews the fused score reached a ROC-AUC of 0.952 and a recall of 0.953, and the two text signals proved complementary, each catching the kind of fake review that the other misses.

Equally important, the project measured where the system should not be trusted: the transfer of hotel-trained deception cues to product reviews is unverified, very short genuine reviews are flagged too often, and the ChatGPT result rests on a small sample. These boundaries are reported to the user and in this report, in line with the principle that an authenticity score should support human judgement rather than replace it.

6.2 FUTURE WORK

The evaluation points to the following directions for further work.

- **Deploy the evaluated safeguards.** Promote the abstention rule for reviews under 120 characters and the domain gate for the deception signal after weighing their coverage cost, so that the platform declines to score where it is known to be unreliable.
- **Product-domain deception data.** Build or obtain a labelled corpus of human-written fake product reviews, so that the deception signal can be validated, retrained and calibrated for the target domain.
- **Broader AI-text evaluation.** Evaluate on a larger and more varied set of reviews from current language models, and compare transformer-based detectors with the present linear model under the same cross-generator protocol.
- **Calibrated fusion.** Learn the signal weights and band limits from labelled data and calibrate the probabilities, so that the score can be interpreted more precisely.
- **Coordinated campaign detection.** Extend behaviour analysis across reviewers, for example by linking accounts that post similar text or target the same products within the same time window.
- **Production readiness.** Add authentication, role-based access and rate limiting, a browser-based CSV upload page and database-side aggregation for large review volumes.

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